SIMULATOR
EXAMINATION SCENARIO GUIDE

SCENARIO TITLE:	14-01 NRC ESG-4
SCENARIO NUMBER:	14-01 NRC ESG-4
EFFECTIVE DATE:	See Approval Dates below
EXPECTED DURATION:	65 minutes
REVISION NUMBER:	00
PROGRAM:	[ ] **L.O. REQUAL**
	[X] **INITIAL LICENSE**
	[ ] **STA**
	[ ] **OTHER** ________________

Revision Summary

New Issue for 14-01 NRC Exam

PREPARED BY:	G Gauding	8-31-15
	Lead Regulatory Exam Author	**Date**
APPROVED BY:	Dal Williams	10/7/15
	Operations Training Manager	**Date**
APPROVED BY:	MW (MDG)	10 9 15
	Facility Representative	**Date**

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

A. Given the unit at 100% power, take corrective action for a loss of CCW IAW AB.CC-0001.

B. Given indication of a loss or malfunction of a safety related plant cooling water system, DIRECT the response to the loss or malfunction in accordance with the approved station procedures.

C. Given the indication of excessive steam flow, perform actions as the nuclear control operator to RESPOND to excessive flow in accordance with S1/S2.OP-AB.STM-0001.

D. Given the indication of excessive steam flow, DIRECT the response to excessive flow in accordance with S1/S2.OP-AB.STM-0001.

E. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.

F. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures

G. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.

H. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.

I. Given the order or indications of a loss of coolant accident (LOCA), complete actions as the nuclear control operator to PERFORM the immediate response to the LOCA in accordance with the approved station procedures.

J. Given the unit with a RCS Leak outside containment, PERFORM actions to isolate the leak, IAW approved station procedures.

K. Given the unit with a RCS Leak outside containment, DIRECT actions to isolate the leak, IAW approved station procedures.

II. MAJOR EVENTS

A. CCW pump trip

B. Steam leak outside containment

C. 2 stuck rods post trip

D. LOCA outside containment

III. SCENARIO SUMMARY

A. The crew will take the watch with Unit 2 at 100% power, EOL. Personnel are in containment to investigate a reduced containment sump pump run interval.

B. Shortly after the crew assumes the watch, the crew in containment will report the source of the leakage was a leaking CFCU SW drain valve, which has been tightened and capped to isolate the leak. The crew will exit containment, but not be able to close the exterior containment door. After the CRS identifies Tech Specs associated with this problem, a Maintenance Supervisor will find a small flashlight preventing airlock door closure, report no damage to the airlock door or seal, and close the airlock door.

C. 22 Component Cooling water pump will trip. The standby CCW pump (23) will not automatically start. The crew will manually start 23 CCW pump. The CRS will ensure the actions of appropriate AB's have been taken.

D. A steam leak outside containment will occur. The crew will enter S2.OP-AB.STM-0001. As the steam leak worsens, the crew will trip the Rx and initiate a MSLI to isolate the steam leak.

E. Once in EOP-TRIP-2, Reactor Trip Response following the immediate actions of EOP-TRIP-1, Reactor Trip or Safety Injection, operators will verify adequate AFW flow and secure SGFPs. Operators will borate the RCS in response to 2 stuck out rods.

F. Once RCS boration is established, a LOCA outside containment occurs on 22 RHR pump piping with check valve failures. Operators will diagnose the loss of PZR level, ECCS Accumulator losing level and pressure, RHR sump pump runs and 2R41D rising radiation, and initiate a Safety Injection based on TRIP-2 Continuous Action Summary, and return to TRIP-1.

G. Operators will perform TRIP-1 and transition to EOP-LOCA-6, LOCA Outside Containment.

H. Once in LOCA-6, operators will isolate the leakage path to 22 RHR loop.

I. The scenario will terminate when the transition from LOCA-6 has occurred.

IV. INITIAL CONDITIONS

____ Presnapped IC – 234

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
____ 1	VC1and VC4 C/T
____ 2	RCPs (SELF CHECK)
____ 3	RTBs (SELF CHECK)
____ 4	MS167s (SELF CHECK)
____ 5	500 KV SWYD (SELF CHECK)
____ 6	SGFP Trip (SELF CHECK)
____ 7	23 CV PP (SELF CHECK)
____ 8	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description	
	1	EVENT ACTION:	kaq09tc7 // 21CA330 CNTMNT CONTROL AIR-2A HE
		COMMAND:	DMF VL0374
		PURPOSE:	<update as needed>
	3	EVENT ACTION:	kaq10tc7 // 22CA330 CNTMNT CONTROL AIR-2B HE
		COMMAND:	DMF VL0375
		PURPOSE:	<update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
______ 01	CC0361C 23 COMPONENT COOLING PUMP Fails to Start on Low Pressure	N/A	N/A	N/A	N/A	
______ 02	CC0172B 22 COMPONENT COOLING PUMP TRIP	N/A	N/A	N/A	RT-5	
______ 03	MS0091r Main Steam Header Leak Outside Containment	N/A	N/A	N/A	RT-7	1
______ 04	RD0064 ANY ROD(S) FAILS TO TRIP (STUCK	N/A	N/A	N/A	N/A	45
______ 05	VL0374 21CA330 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	100
______ 06	VL0375 22CA330 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	100
______ 07	SJ0312A CL LEG INJ LINE CHK VALVE 24SJ56 (RCS SIDE) LEAKS (use with SJ0312B)	N/A	0	00:10:00	RT-9	8
______ 08	SJ0312B CL LEG INJ LINE CHK VALVE 24SJ43 (RHR SIDE) LEAKS	N/A	0	00:10:00	RT-9	8
______ 09	RH0300B 22 RHR LEAK AFTER HX	N/A	N/A	N/A	RT-9	500

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
______ 01	RC05A RCS SYSTEM , BORON CONC RESET	N/A	N/A	N/A	N/A	27.7
______ 02	CT02D EL 103 AIRLOCK INT DOOR CLOSED	N/A	N/A	N/A	RT-1	NO
______ 03	CT01D EL 103 AIRLOCK EXT DOOR CLOSED	N/A	N/A	N/A	RT-3	NO

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/ Severity
______ 01	A701 B DI TRAIN 'B' - SI OPERATE KEYSWITCH	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:	
	Description
____ 1.	

V. SEQUENCE OF EVENTS

A. State shift job assignments and review scenario objectives.

B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).

C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".

D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. 100' elevation containment airlock door failure to close			
Simulator Operator: Call control room as Zach Crawford and report that your crew identified a leaking drain valve (24SW269) on the outlet of 24 CFCU which did not have its drain cap installed. Your crew tightened the valve to stop the leak, put the valve cap on, and your crew will be exiting containment. After report, Insert **RT-1** to open interior containment airlock door. **REMOTE: CT02D EL 103 AIRLOCK INT DOOR CLOSED Final Value: NO** 15 seconds later, MODIFY **REMOTE: CT02D to YES.** Then insert **RT-3** to open airlock exterior door. **REMOTE: CT01D EL 103 AIRLOCK EXT DOOR CLOSED Final Value: NO**			
	RO announces expected OHA C-46 PERSONNEL ACCESS DOOR OPEN.		
Role Play: After 3 minutes with airlock door open, call control room			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
as Zach Crawford and report that the door closing handwheel will not fully close the 100' airlock exterior door and you need Mechanical Maintenance to come look at door. Specifically, the containment door remains approximately 2 inches open.			
	CRS contact maintenance.		
	CRS enters TSAS 3.6.1.1.b and 3.6.1.3.a.		
Role Play: <u>After</u> the CRS has determined applicable Tech Specs, **MODIFY REMOTE CT01D to YES,** then call as Maintenance Supervisor and report the reason the door would not close was a small flashlight was blocking its path. You have inspected the door seal and there is no damage.			
	RO reports OHA C-46 has cleared.		
2. 22 CCW pump trip with failure of 23 CCW pump to auto start.			
Simulator Operator: Insert <u>**RT-5**</u> on direction from Lead Evaluator. **MALF: CC0172B 22 CCW pump trip**			
	RO reports trip of 22 CCW pump and standby		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	pump did not auto start.		
	CRS directs RO to manually start 23 CCW pump.		
	RO manually starts 23 CCW pump and reports clearing of all alarms associated with the CCW pump trip.		
	CRS directs RO/PO to validate alarms received with ARP guidance.		
Note: CCW system alarms will annunciate during crew response, but all alarms will clear upon restoration of system pressure.			
	RO/PO report that alarms received were consistent with low CCW system pressure, and that the restoration of system pressure was expected to clear all those alarms.		
	RO verifies RCP CCW cooled parameters trending to normal values.		
	CRS may refer to S2.OP-AB.CC-0001, Component Cooling Abnormality, and S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality, to verify entry is not required.		
Note: Entry into these 2 AB's may be performed to verify the actions taken to restore CCW system pressure corrected the entry conditions for the AB's.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS dispatches operators to investigate 22 CCW pump trip.		
Role Play: 2 minutes after being dispatched to check 22 CCW pump breaker, report OC trip is present on 22 CCW pump. If dispatched to check CCW pumps report they look normal.			
	CRS enters TSAS 3.7.3 based on not having 2 operable CCW loops.		
Proceed to next event at Lead Evaluators direction after Tech Spec call has been made.			
3. Steam leak outside containment / power reduction			
Simulator Operator: Insert **RT-7** at Lead Evaluators direction. **MALF: MS0091r, Main Steam Header Leak Outside Containment** **Severity: 1**			
	RO reports rising power and lowering Tavg.		
	RO reports no indication of RCS leak.		
	CRS enters S2.OP-AB.STM-0001, Excessive Steam Flow.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS directs initiation of AB.STM CAS.		
	CRS notifies Emergency Services.		
	PO reports Main turbine is latched.		
	PO reports EHC operating correctly.		
	CRS directs the PO to lower turbine load to lower reactor power ≤ 100% power.		
	PO lowers turbine load at rate directed by CRS to level directed by CRS.		
	RO initiates a boration as directed by the CRS to maintain Tavg on program.		
	PO reports no indication of any MS10 malfunction.		
	PO reports no indication of Main Steam Dump malfunction.		
	CRS dispatches operator to check for leaking Safety Valve if not previously dispatched.		
Role Play: 2 minutes after being dispatched, but **AFTER** load reduction has been performed, call as NEO to report a small steam leak from the top of 23 West Moisture Separator Reheater. You can't see exactly where it is coming from.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: On direction from Lead Evaluator after downpower has been performed, **MODIFY MS0091r to 20%.**			
	PO reports rising steam flows on all SGs.		
	RO reports Rx power rising rapidly.		
	CRS briefs actions to be taken to trip the reactor and isolate the steam leak.		
	RO trips the reactor.		
	RO initiates MSLI.		
	PO reports steam flows have all lowered, and MSLI has isolated source of steam leak.		
	RO performs immediate actions of TRIP-1: - Confirms the reactor trip. - Trips the Main Turbine - Reports at least one 4KV vital bus energized. - Reports SI not initiated, and verifies SI not required.		
	CRS / RO verify immediate actions and SI not required.		
	CRS transitions to EOP-TRIP-2, Reactor Trip Response.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO makes page announcement.		
	CRS directs implementation of the ECG.		
	PO reports total AFW flow is >22E4 lbm/hr.		
	PO stops both SGFPs.		
	PO maintains 22E4 lbm/hr total AFW flow until at least 1 SG NR level is >9%, and maintains SG NR levels 9-33%.		
	RO reports RCPs in service and Tavg trending to 547°.		
	RO reports both Rx RTBs are open.		
	PO ensures 21-24BF19 and 21-24BF40 are shut, and shuts 21-24BF22.		
	RO reports 2 control rods have failed to fully insert.		
	RO starts a BAT pump in fast speed.		
	RO opens 2CV175.		
	RO shuts 21 and 22 CV160s.		
	RO controls charging flow to maintain >87 gpm.		
	RO reports rapid boration flow is established.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
CT #1: (TRIP-2) Initiate Rapid Boration for two or more control rods not fully inserted following a Reactor Trip prior to exiting TRIP-2. **SAT_____ UNSAT_____**			
	CRS determines 70 minutes of boration flow is required for the 2 stuck out rods, and assigns a crew member responsibility for tracking the 70 minutes.		
Simulator Operator: Insert RT-9 on direction from Lead Evaluator after rapid boration has been established. **MALF: RH0300B 22 RHR leak after HX** **Final Value: 500** **MALF: SJ0312A CL Leg Inj Line Chk Valve 24SJ56 fails** **Final Value: 8** **Ramp: 10 minutes** **MALF: SJ0312B CL Leg Inj Line Chk Valve 24SJ43 fails** **Final Value: 8** **Ramp: 10 minutes**			
	RO reports 24 ECCS Accumulator level low		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	alarm, followed shortly by the Low Pressure alarm.		
	RO reports RCS pressure dropping slowly.		
	RO checks containment conditions and reports they are stable.		
	RO reports PZR level lowering and RHR sump pump starts.		
	RO raises charging flow to stabilize PZR level.		
	RO reports PZR level cannot be maintained stable with elevated charging flow.		
	CRS directs RO to initiate SI based on CAS of TRIP-2.		
	RO attempts to manually initiates SI, and reports SI will not initiate from Train B. RO initiates SI successfully from Train A.		
	CRS transitions to TRIP-1.		
	CRS/RO verify TRIP-1 immediate actions complete.		
	RO announces OHA C34, 22 RHR SUMP OVRFLO.		
	PO reports SEC loading is not complete for energized vital busses.		
	PO reports all available equipment started.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports 21 and 22 AFW pumps are running.		
	PO reports valve groups in Table B are in safeguards position.		
	RO reports 21 and 22CA330's are NOT shut, and shuts 21 and 22CA330.		
Simulator Operator: Ensure **ET-1** and **ET-3** are true when 21CA330 and 22CA330 close PBs are depressed respectively. This deletes the malfunctions and allows valves to shut.			
	RO reports containment pressure has remained < 15 psig.		
	RO reports no high steam flow condition is indicated on 2RP4.		
	CRS directs SM to implement the ECG.		
	RO reports all 4KV vital busses are energized.		
	RO runs two Switchgear Supply fans and one Switchgear Exhaust fan.		
	RO verifies 2 CCW pumps running.		
	CRS reads LOCA-3 transition CAS.		
	RO reports correct ECCS pumps injecting for		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	current RCS pressure.		
	PO verifies AFW flow and SG NR level status.		
	RO reports all RCPs running and MS10's controlling RCS Tave.		
	RO reports trip breaker and PORV status.		
	RO reports RCPs are in service, and spray valves status.		
	RO reports RCS pressure is >1350 psig.		
	PO reports no indications of any faulted SGs.		
	PO reports no indications of any ruptured SGs.		
	RO reports no radiation monitors indicate a LOCA in containment.		
	RO reports containment pressure is < 4 psig.		
	RO reports containment sump level is <46%.		
	RO reports subcooling is >0°F.		
	PO reports SG NR levels >9%.		
	RO reports RCS pressure is dropping slowly.		
Simulator Operator: If RO reports stable RCS pressure, then increase the size of the RHR leak to raise			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
the rate at which pressure is lowering, to get the crew to get out of the "do loop" of step 29.			
	STA continues monitoring CFST's.		
	PO verifies AFW flow and SG NR level status.		
	RO reports radiation monitor 2R41D in alarm.		
	CRS sends an operator with Rad Pro support to investigate.		
Note: 5 minutes after being dispatched, report as NEO that the RP Tech made both of you leave the area when his monitor alarmed prior to entering the 55' elevation RHR valve room area.			
	CRS determines cause of high radiation is a LOCA outside containment in 22 RHR pump room based on alarms and plant conditions.		
	CRS transitions to LOCA-6.		
	RO resets SI and Phase A isolation, and reports Phase B isolation not actuated.		
	RO opens 21 and 22CA330's.		
	PO resets all SECs, and reports 230V control centers reset.		
	RO reports 2RH1 and 2RH2 are shut.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO shuts 21RH19 and 22RH19.		
	RO reports RCS pressure is not rising.		
	RO reports 2RH26 is shut, and RCS pressure is not rising.		
	RO verifies 21RH29 in auto.		
	PO removes lockout from 21SJ49.		
	RO shuts 21SJ49.		
	RO reports RCS pressure is not rising.		
	RO opens 21SJ49.		
	RO verifies 22RH29 in auto.		
	PO removes lockout from 22SJ49.		
	RO shuts 22SJ49.		
	RO reports RCS pressure is rising.		
CT#2 (CT-32) Isolate LOCA outside containment before transition out of LOCA-6. **SAT____ UNSAT____**			
	RO stops 22 RHR pump.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS transitions to LOCA-1.		
Lead Examiner terminate the scenario when the transition to LOCA-1 is announced.			

VI. SCENARIO REFERENCES

A. Alarm Response Procedures (Various)

B. Technical Specifications

C. Emergency Plan (ECG)

D. OP-AA-101-111-1003, Use of Procedures

E. S2.OP-AB.CC-0001, Component Cooling System Abnormality

F. S2.OP-AB.STM-0001, Excessive Steam Flow

G. 2-EOP-TRIP-1, Reactor Trip or Safety Injection

H. 2-EOP-TRIP-2, Reactor Trip Response

I. 1-EOP-LOCA-1, Loss of Reactor Coolant

J. 2-EOP-LOCA-6, LOCA Outside Containment

ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY

MODE: 1 POWER: 100 RCS BORON: 100 MWe 1220

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

Personnel are in containment investigating reduced Containment Sump Pump run interval.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

None

SECONDARY:

None

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

None

ATTACHMENT 2

SIMULATOR READY FOR TRAINING CHECKLIST

____ 1. Verify simulator is in "TRAIN" Load

____ 2. Simulator is in RUN

____ 3. Overhead Annunciator Horns ON

____ 4. All required computer terminals in operation

____ 5. Simulator clocks synchronized

____ 6. All tagged equipment properly secured and documented

____ 7. TSAS Status Board up-to-date

____ 8. Shift manning sheet available

____ 9. Procedures in progress open and signed-off to proper step

____ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced

____ 11. Required chart recorders advanced and ON (proper paper installed)

____ 12. All printers have adequate paper AND functional ribbon

13. Required procedures clean

____ 14. Multiple color procedure pens available

____ 15. Required keys available

____ 16. Simulator cleared of unauthorized material/personnel

____ 17. All charts advanced to clean traces and chart recorders are on.

____ 18. Rod step counters correct (channel check) and reset as necessary

____ 19. Exam security set for simulator

____ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter

____ with Baseline Data filled out

____ 21. Shift logs available if required

____ 22. Recording Media available (if applicable)

____ 23. Ensure ECG classification is correct

____ 24. Reference verification performed with required documents available

____ 25. Verify phones disconnected from plant after drill.

____ 26. Verify ECG paperwork is marked "Training Use Only" and is current revision.

____ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3

CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.
The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...

- degradation of any barrier to fission product release
- degraded emergency core cooling system (ECCS) or emergency power capacity
- a violation of a safety limit
- a violation of the facility license condition
- incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
- a significant reduction of safety margin beyond that irreparably introduced by the scenario

II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...

- effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
- recognize a failure or an incorrect automatic actuation of an ESF system or component.
- take one or more actions that would prevent a challenge to plant safety.
- prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 14-01 NRC ESG-4 **REVIEWER: R Blose**

Initials		Qualitative Attributes
BB	1.	The scenario has clearly stated objectives in the scenario.
BB	2.	The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
BB	3.	The scenario consists mostly of related events.
BB	4.	Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
BB	5.	No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
BB	6.	The events are valid with regard to physics and thermodynamics.
BB	7.	Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
BB	8.	The simulator modeling is not altered.
BB	9.	All crew competencies can be evaluated.
BB	10.	The scenario has been validated.
BB	11.	If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
BB	12.	ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	4	Malfunctions after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5

ESG CRITICAL TASKS

14-01 NRC ESG-4

CT #1 (TRIP-2) (FSAR 4.2.3.2.1) Initiate Rapid Boration for two or more control rods not fully inserted following a Reactor Trip prior to exiting TRIP-2.

BASIS: Core Shutdown Margin (SDM) is only assured with a maximum of one control rod not fully inserted. With two or more control rods not inserted, the shutdown reactivity margin must be made up through emergency boration to account for the reactivity of the stuck rods.

CT #2: (CT-32) Isolate LOCA outside containment before transition out of LOCA-6.

BASIS: Failure to isolate a LOCA outside containment (that can be isolated) degrades containment integrity beyond the level of degradation irreparably introduced by the postulated conditions. It also constitutes misoperation or incorrect crew performance that fails to prevent "degradation of any barrier to fission product release" and eventually leads to "...degraded emergency core cooling (ECCS)...capacity."

ATTACHMENT 6

PRA-ESG RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

Y/N	Event	Y/N	Event
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	Y	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
N	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

Y/N	COMPONENT, SYSTEM, OR TRAIN	Y/N	COMPONENT, SYSTEM, OR TRAIN
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	N	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

Y/N	OPERATOR ACTION
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.